

Hassan Nauman

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EDUCATION

University of Manitoba

Bachelor of Science in Computer Science

Winnipeg, MB

May 2019 – Dec 2024

PROJECTS

Subscription Tracker Android Application / Java, HSQLDB, XML, Junit, Bootstrap, HTML, CSS, JS, Git

- Developed a **Full Stack Subscription Management App in Android Studio**, leveraging **HSQLDB** for secure user data storage and advanced analytics, **achieving a 60% reduction in user spending** by optimizing subscription tracking and management.
- Engineered a robust **3-tier architecture**, integrating **Front-End (XML)**, **Logic Layer (Java)**, and **Database (HSQLDB)**, which **ensured a well-structured and efficient system design**, enhancing application performance and scalability.
- Implemented **Agile methodologies** and utilized **Git for efficient code management**, facilitating adaptive development and ensuring the timely delivery of the project through streamlined processes and version control.
- Developed a promotional **website for the application** using **Bootstrap, HTML, and JavaScript**, enhancing user interaction and feature presentation, which facilitated a deeper understanding with the app's functionalities.
- Conducted comprehensive **software testing**, including unit, integration, and system tests using **JUnit**, increasing **software quality and reliability by 40%** through robust coverage of all edge cases.

Urban Data Analytics System for Winnipeg / Java, SQL, Python, Git

- Developed an **SQL-based analytics tool for urban planners**, providing real-time access to Winnipeg's data on trees, parks, and neighborhoods. This tool **enhanced data analysis efficiency by 30%**, enabling quicker and more informed decisions on environmental and urban planning through the advanced visualization of tree species distribution and park assets.
- Crafted **Python scripts** to manipulate data from the **City of Winnipeg**, converting it into a **SQL-readable format** for query execution with **Java**, streamlining the **data integration** process and **enhancing the efficiency** of database operations.

Artificial Intelligence Research Paper

- Authored a **research paper** under **Dr. John Anderson's** supervision, **detailing neural network optimization's evolution and mechanisms**, with a focus on supervised training and backward propagation. **Investigated Stochastic Gradient Descent (SGD) and the Adam optimizer**, conducting a comparative analysis to highlight their strengths and weaknesses, contributing to a deeper understanding and application of neural network optimization strategies.
- Investigated a **hybrid optimization strategy** combining SGD and Adam's benefits, applied to Convolutional Neural Network (CNN) training. This approach demonstrated a **15% increase in validation accuracy**, showcasing the potential of hybrid methods to enhance efficiency and adaptability in neural network optimization, contributing valuable insights to the field.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, XML, MongoDB

Frameworks: React, ExpressJS, Bootstrap, Node.js, JUnit, Rest API, Fast API

Developer Tools: Git, VS Code, Visual Studio, IntelliJ, Eclipse, Jupyter Notebook, PyCharm,

Libraries: pandas, NumPy, Matplotlib

EXPERIENCE

Car Return Associate

Jun 2023 – Present

Avis Car Rental, Winnipeg Richardson International Airport

Winnipeg, MB

- Performed comprehensive vehicle inspections to identify and document any damages with precision.
- Created detailed damage reports by referencing extensive records of previous contracts.

Sales Associate

Sep 2022 – Dec. 2022

TELUS Mobility

Winnipeg, MB

- Achieved sales targets by effectively promoting telecommunications products and services to a diverse client base.
- Achieved top-seller of the month title by surpassing the sales target, resulting in 40% increase in sales performance.